

BLAKE HARMS

SENIOR SOFTWARE ENGINEER & AI ENTHUSIAST

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🌐 harms-haus.github.io/resume

Core Languages:

TypeScript

C#

JavaScript

Python

SQL

Bash/Zsh

Rust

Golang

Frontend:

Angular

HTML5/CSS3

WPF

React

SASS

Backend:

Node.js

REST API

Vector DB

Microservices

Redis

AI Systems:

RAG Architect

Prompt Eng

Cursor

LangChain

Copilot

Local LLMs

DevOps:

Docker

CI/CD

Git

Azure DevOps

Linux Admin

Architecture:

UX Design

Home Auto

Tech Lead

Agile

// PROFESSIONAL SUMMARY

Full-stack software engineer with 8+ years experience (Emerson, Wood), specializing in high-performance C# simulations, Angular-based web platforms, and resilient DevOps infrastructure. Successfully led a 4-year initiative to modernize mission-critical desktop systems into cloud-ready Angular environments, reducing technical debt by 60%. Expert in architecting integration layers, implementing automated testing cultures, and driving safe AI/Copilot adoption within engineering workflows.

// EXPERIENCE

Senior Software Engineer

APR 2017 – PRESENT

Emerson

- **Framework Modernization:** Architected the transition of a legacy C# WPF desktop framework into an Angular/TypeScript web platform, achieving near-perfect feature parity over a 4-year solo project.
- **Strategic Tech Debt Resolution:** Refactored massive legacy codebases to resolve ~60% of technical debt; introduced domain-driven architecture and rigorous testing (Vitest/Jest).
- **Engineering Leadership:** Oversaw a modernization squad; developed reusable integration and design systems leveraging Angular, TypeScript, and modern CSS frameworks.
- **DevOps Modernization:** Led full SVN-to-Git migration across complex monorepos; implemented Docker-based containerization and automated Azure CI/CD pipelines.
- **AI Integration & Governance:** Established best practices for GitHub Copilot usage via custom tooling and training. Championed modern PR review workflows.
- **Systems Performance:** Developed high-throughput C# simulation engines for industrial engineering; iterated on UX via deep user research.

Wood

- › **Industrial Software Design:** Built WPF MVVM applications for specialized Oil & Gas fluid dynamics simulation tooling.
- › **Operational Efficiency:** Streamlined complex engineering configurations through intuitive UI workflows, drastically reducing time-to-result for simulation runs.

// AI & SYSTEMS EXPLORATION

Jester

LLM FRAMEWORK

Advanced context-steered storytelling framework built on hybrid GraphRAG. Maintains 40+ interconnected narratives with high consistency using 10,000 lines of custom markdown. Context is selectively included based on the current stage of the story generation. Zero code.

GraphRAG • Context Selection • LLM Multi-Agent System

pi-coding-agent Extensions

AGENTIC DEVELOPER TOOLING

A curated collection of extensions for the [pi coding agent](#) — an agentic coding workflow including deterministic workflow steps, a phased task board with dependency tracking and status gating (ready → implementing → reviewing → done), parallel sub-agent orchestration, and more. Built in TypeScript using pi's extension SDK and published to npm. github.com/harms-haus/pi-extensions

TypeScript • AI Agent System • CLI • Developer Tooling • npm

AI Buddy

KIDS' VOICE ASSISTANT

Privacy-focused voice assistant for my kids. Uses Mastra (agent orchestration), Whisper (STT), and Kokoro (TTS) over Home Assistant's ESPHome voice-assistant pipeline (Wyoming protocol) to inspire curiosity, play kid-friendly music, and control lighting in their rooms. github.com/harms-haus/ai-buddy

TypeScript • Python • Mastra • Whisper • Kokoro • Home Assistant • ESPHome

Ymir

WORKSPACE MANAGER

Rust (Tauri)/React web and desktop app to manage terminal tabs, browser tabs, and editors in workspaces. Features worktree support for sub-workspaces, secure websockets, and both web/Tauri access. github.com/harms-haus/ymir

Rust • Tauri • React • WebSockets • Workspace Management

// EDUCATION

B.S. Computer Science

University of Houston
Graduated 2012

Minor in Mathematics